

**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**

Paper Code : MCAN-102 Relational Database Management System

UPID : 001608

Time Allotted : 3 Hours

Full Marks : 70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) During the _____ phase, the local variables that hold the write operations are copied to the database.
- (II) The dependency rules specified by the database designer are known as _____.
- (III) _____ is a predicate that we expect the database to always satisfy.
- (IV) What is the format of entering date into a database while inserting data into it?
- (V) To select a tuple from a relational database table, the symbol used in relational algebra is _____.
- (VI) Which of the systems use RDMS?
- (VII) What is the cardinality of a table of 50 rows and 5 columns?
- (VIII) Data about data is known as _____.
- (IX) DML is provided for what purpose?
- (X) Grant and revoke are _____ statements.
- (XI) Which command cannot be used to modify the data in a database?
- (XII) A transaction for which all committed changes are permanent is called _____?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

- 2. Briefly explain different JOIN operation with example. [5]
- 3. Explain the three-tier architecture. What is the difference between two-tier architecture and three-tier architecture? [5]
- 4. Define Optimistic Concurrency Control with example. [5]
- 5. Define Entity. Explain Strong and Weak Entity Set. [5]
- 6. What is primary indexing and secondary indexing? [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

- 7. How does a B-tree differ from a B+ tree ? Outline an algorithm for insertion of a record in a B+ tree. [15]
Construct a B+ tree for the following set of key values under the assumption that the number of key values that fit in a node is 3. Key values are (3, 10, 12, 14, 29, 38, 45, 55, 60, 68). Use your algorithm to show the steps involved for inserting 11 and 30.
- 8. Consider the relational database, where the primary keys are underlined. [15]
Employee(ename,street,city)
Works(ename,company_name,salary)
Company(company_name,city)
Manages(ename,manager_name)\
Write the query expressions for each of the following queries in special language:
 - a. Find the names of all employees who work for the First Bank Corporation.
 - b. Find the names of all employees who live in the same city and on the same street as do their managers.
 - c. Display the employee name, employee number for all employees who earn more than average salary.
 - d. Display the employee number, employee name for all employees who work in a department with any employee whose name contains a 'T'.
- 9. What is the concept of sub-query in terms of SQL? [15]
What is the use of DROP command and what are the differences between DROP, TRUNCATE and DELETE commands? What is Correlated Subquery in DBMS? What is the main difference between UNION and UNION ALL?

10. What is the difference between DBMS and RDBMS?

[15]

What is Entity Integrity and Referential Integrity?

11. Why ER model is used? Benefits of ER modeling.

[15]

Find the minimum number of tables required to represent the given ER diagram in relational model.

Applying the rules that we have learnt, minimum 6 tables will be required-

- Account (Ac_no , Balance , b_name)
- Branch (b_name , b_city , Assets)
- Loan (L_no , Amt , b_name)
- Borrower (C_name , L_no)
- Customer (C_name , C_street , C_city)
- Depositor (C_name , Ac_no)

*** END OF PAPER ***